

Matrix Computations

A Useful Proof Technique

Hongyuan Zha

School of Data Science
The Chinese University of Hong Kong, Shenzhen

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An optimization problem

- To prove that

$$\eta = \max_x f(x)$$

we follow the following two steps

- 1 We first prove

$$\forall x, \quad f(x) \leq \eta \quad \implies \max_x f(x) \leq \eta$$

- 2 Then we show

$$\exists x^*, \quad f(x^*) \geq \eta \implies \max_x f(x) \geq \eta$$

Example: matrix 1-norm

- 1-norms: let $A = [a_1, \dots, a_n]$, then

$$\|A\|_1 = \max_{1 \leq i \leq n} \|a_i\|_1$$

- Let

$$\eta = \max_{1 \leq i \leq n} \|a_i\|_1$$

We need to show that

$$\eta = \max_{x \neq 0} \frac{\|Ax\|_1}{\|x\|_1}$$

Example: matrix 1-norm

- For step one:

$$\begin{aligned}\|Ax\|_1 &= \|x_1 a_1 + \cdots + x_n a_n\|_1 \\ &\leq |x_1| \|a_1\|_1 + \cdots + |x_n| \|a_n\|_1 \\ &\leq \eta(|x_1| + \cdots + |x_n|) \\ &= \eta \|x\|_1\end{aligned}$$

- For step two: let

$$\|a_k\|_1 = \max_{1 \leq i \leq n} \|a_i\|_1 = \eta$$

and choose $x^* = e_k$, the k -th column of the identity matrix. Then

$$\|x^*\|_1 = 1, \quad \|Ax^*\|_1 = \|a_k\|_1 = \eta \|x^*\|_1$$

Example: matrix 2-norm and Frobenius norm

- We want to prove the following

$$\|A\|_2 = \max_{B \neq 0} \frac{\|AB\|_F}{\|B\|_F}$$

where A and B are matrices

- For step one: we need to show

$$\|AB\|_F \leq \|A\|_2 \|B\|_F$$

to this end, partition B column-wise $B = [b_1, \dots, b_p]$. Then

$$\|AB\|_F^2 = \|[Ab_1, \dots, Ab_p]\|_F^2 = \sum_{i=1}^p \|Ab_i\|_2^2 \leq \sum_{i=1}^p \|A\|_2^2 \|b_i\|_2^2 = \|A\|_2^2 \|B\|_F^2$$

Example: matrix 2-norm and Frobenius norm

- For step two: let u be such that

$$\|Au\|_2 = \|A\|_2\|u\|_2$$

and $B = [u, 0, \dots, 0]$. Then

$$\|B\|_F = \|u\|_2,$$

and

$$\|AB\|_F = \|[Au, 0, \dots, 0]\|_F = \|Au\|_2 = \|A\|_2\|u\|_2 = \|A\|_2\|B\|_F$$